Paper I-B — Leaf-Spring Bent-Tab Adaptive Tray (HyTray-B): No-Separate-Spring Geometric Leaf-Spring Flap with Distributed Stress and Criss-Cross Option

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Invention Disclosure / Provisional Specification Draft — HyTray Series HT-01B

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Abstract

A cross-flow distillation tray is disclosed in which vapour passages are closed by **lanced-and-formed bent tabs**: each tab is cut on three sides and bent up at a generous radius (4–8 times deck thickness) over a formed leg, so the geometry of the tab itself acts as a leaf spring — the plate material over the spring leg, not a separate spring component, provides the return force. A parallel lance adjacent to the opening sets the effective spring-leg length. **No separate spring part exists.** Spring rate is controlled entirely by leg length L, bend radius r_b and deck thickness t: $k_{spring} = Et^3w/(4L^3)$. Families of differing spring rates give the same continuous open-area schedule $A_o(U_s)$ as HT-01A with materially better fatigue (stress distributed along the leg rather than concentrated at a thin pivot land). The criss-cross interlaced installation of HT-01A applies equally to this embodiment. Importantly: this design has **zero additional moving or corrodible parts beyond the deck plate itself**.

Status note: calibration of spring rate, crack ΔP and fatigue life against the parametric equations below is required before performance numbers are asserted. No separate spring components are used; concerns about spring corrosion, ductility and fatigue do not apply to this embodiment.

1 Why Not Actual Springs?

A coil or torsion spring in distillation service would face: (i) fatigue in cyclic loading at high frequency — spring steel has a lower endurance ratio than austenitic SS; (ii) stress-corrosion cracking in H_2S/amine/HCl; (iii) spring-back loss (set) at elevated temperature; (iv) one more loose part that can land in the downcomer or pump suction. All of these are resolved by the geometric leaf spring: the deck plate IS the spring. Same material as the rest of the tray, no additional corrosion surface, no additional failure mode.

2 Mechanical Character and Fatigue

The spring leg of length L, width w, thickness t acts as a cantilever. For tip deflection $\delta = L \sin\theta$:

$$\sigma_a = \frac{3Et\,\delta}{2L^2} = \frac{3Et\sin\theta}{2L}.$$

Comparing to the coined hinge at the same θ: the hinge concentrates the same rotation in $\ell_h \ll L$, giving stress $\propto t/\ell_h$; the leaf spring gives stress $\propto t/L$. For $L = 35\,\text{mm}$, $\ell_h = 3\,\text{mm}$: stress ratio

$\approx \ell_h/L = 0.086$, i.e. the leaf spring sees roughly 12× lower bending stress for the same opening angle. Staying below the SS endurance limit $\sigma_e \approx 160\,\mathrm{MPa}$ with $K_f = 1.2$ (smooth-formed bend) requires $L \geq 3Et\sin\theta/2\sigma_e K_f$. At $t = 2.5\,\mathrm{mm}$, $\theta = 24°$: $L_{min} \approx 15\,\mathrm{mm}$; the preferred 30–40 mm leg provides margin of > 2.

Spring rate $k = Et^3w/(4L^3)$. The three families are differentiated by leg length alone (simpler tooling than changing thickness):

Soft: $L = 40\,\mathrm{mm} \Rightarrow k_A$; Med: $L = 32\,\mathrm{mm} \Rightarrow k_B = k_A(40/32)^3 = 1.95k_A$; Stiff: $L = 25\,\mathrm{mm} \Rightarrow k_C = 4.10k_A$.

Cracking pressure for lip area A_{lip}: $\Delta P_{cr} = k\delta_{open}/A_{lip}$; the ratio above gives the required 1:2:4 crack-pressure spread. No coins, no coiling, no heat treatment — one press form, three leg-length programmes in the stamping die.

3 Brief Description of the Drawings

Figure 1: (a) cross-section showing the long elastic leg distributing bending stress (low everywhere, quantified by the bending-stress profile overlay), the parallel lance, and the large-radius bend; (b) plan view of one tray showing interspersed spring-rate families identified by leg length.

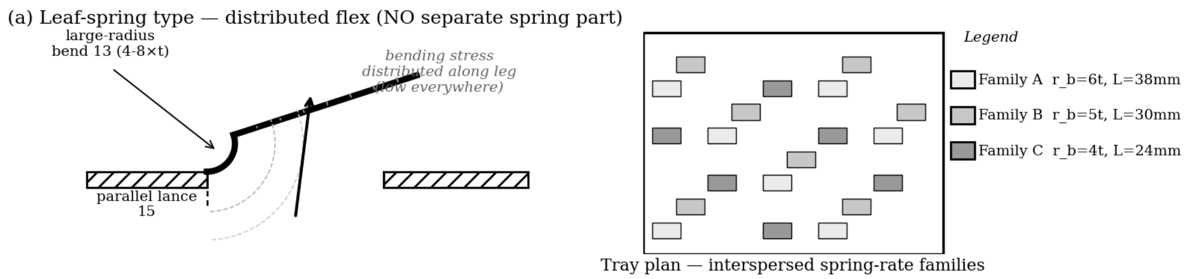

Figure 1: (a) Geometric leaf-spring: bending stress distributed along leg (grey tick marks show low stress gradient); large-radius bend *(13)*; parallel lance *(15)*. No separate spring component. (b) Three leg-length families interspersed on the deck plan — Family A longest (softest), C shortest (stiffest).

4 Criss-Cross — identical to HT-01A

The criss-cross interlaced installation (Claim 4 of HT-01A) applies without modification to this embodiment; the spring-rate family is the layout dimension that gets mirrored. The same vapour-zigzag benefit applies.

5 Design Parameter Envelope

6 Claims (draft)

Claim 1. A cross-flow contacting tray comprising a deck plate having a plurality of flap valves, each flap being formed by lancing three sides of a rectangle in the deck plate and bending the freed tongue over a radius of 4–8 times the deck thickness to form an integral leaf spring, the return force provided entirely by the bending stiffness of the plate material over the spring leg, with no separate spring component.

Claim 2. The tray of claim 1, wherein the flaps are provided in at least two spring-rate families differing in leg length L, the spring rate being $k = Et^3w/4L^3$, such that the families crack open

Parameter	Claimed band	Preferred
Bend radius / deck thickness r_b/t	4–8	6
Spring leg length L	20–50 mm	25/32/40 (stiff/med/soft)
Spring leg width w	10–25 mm	14 mm
Spring rate $k = Et^3w/4L^3$	computed per family	3-family ratio 1:2:4
Crack ΔP	1.0–7 mbar	1.2/2.4/4.8 mbar
Max bending stress σ_a at θ_{max}	$< \sigma_e/K_f$	$<$ 130 MPa
Parallel lance clearance g	0.15–0.30 mm	0.22 mm
$A_{o,min}$ (shear gap)	3–6 %	4 %
Corrodible/moving parts beyond deck	**zero**	zero

sequentially and the aggregate open area varies continuously between 3–6% and 14–18% of bubbling area.

Claim 3. The tray of claim 1, wherein a parallel lance adjacent to the opening sets the effective spring-leg length, and the peak bending stress at maximum lift angle satisfies $\sigma_a = 3Et\sin\theta/2L < \sigma_e/K_f$ over a fatigue life of at least 10^8 cycles.

Claim 4. The tray of claim 1, wherein no hole is placed in the bend region, and minimum open area is provided by the perimeter shear gap of the lanced opening.

Claim 5. The tray of claim 1, wherein the device is free of separate springs, captive valves, pins, or any component not integral to the deck plate, eliminating spring corrosion, ductility loss, and spring-set as failure modes.

Claim 6. A column assembly in which alternate trays have their spring-rate family layouts mirror-inverted, as claimed in HT-01A claim 4, applied to trays according to claims 1–5.